Burner

A Swarm of Disposable Agents for Unattributable Web Interaction

V1.0

Abstract

The modern web identifies its users by default. Every interaction leaves a persistent record: a device fingerprint, a session, a pattern of behavior, all bound to a single durable identity that follows a person across services and accumulates into a profile they neither see nor control. This paper presents Burner, a system that breaks the link between a person and their activity on the web. Rather than acting on a site directly, a user delegates a task to an autonomous agent that performs it inside an isolated browser context under a disposable identity. The identity exists only for the duration of the task and is destroyed on completion, leaving no persistent state by which the activity can later be attributed. Tasks larger than a single interaction are distributed across many such agents operating in parallel, coordinated toward the user's intent but unlinkable to one another from the perspective of any external observer. The result is a model in which a user is represented online not by one consistent identity but by a continuous supply of unrelated strangers, none of which can be traced back to the user or to each other. We describe the threat model this addresses, the architecture that realizes it, and the properties and limits of the approach.

1 Introduction

The web was built to be open. In practice it has become an apparatus for observation. The dominant business model of the consumer internet is the collection, retention, and monetization of behavioral data, and the technical infrastructure of the web has been shaped to serve that model. A user who visits a site is fingerprinted, cookied, and correlated. Their actions are logged and joined to prior actions, and over time these records resolve into a detailed profile attached to a stable identity.

This identification is not a consequence of wrongdoing on the user's part. It is the default condition of using the web as a single, persistent self. The longer a person operates under one identity, traversing the same services along recognizable patterns, the more complete the profile that can be assembled about them. The cost of participation is exposure, and that cost is extracted whether or not the user consents to it in any meaningful sense.

Existing privacy tools address parts of this problem but not its center. A VPN conceals network origin but does nothing about the persistent identity a user presents to a service once connected. Private browsing modes clear local state but do not prevent fingerprinting or behavioral correlation. Anti-tracking extensions reduce some third-party collection but leave first-party identification intact.

Each of these narrows the surface of exposure. None of them removes the fundamental vulnerability, which is that the user is a single, consistent, observable entity at the point of interaction.

Burner takes a different approach. Rather than reducing what a persistent identity leaks, it removes the persistent identity altogether. The user does not interact with the web as themselves at all. They delegate, and what the web observes is never the user.

2 Threat Model

We assume an adversary consistent with the ordinary operation of the modern web rather than a targeted attacker. The adversary is any party, first or third, capable of observing interactions with a web service and seeking to attribute those interactions to a persistent identity. This includes the operators of the services a user interacts with, the analytics and advertising infrastructure embedded in those services, and any party able to aggregate records across services.

The adversary's goal is correlation: to determine that distinct interactions originate from the same entity, and to bind that entity to an identity that persists over time. The adversary's tools are the standard instruments of web identification. These include cookies and other client-side storage, device and browser fingerprinting, network-level identifiers such as IP address, and behavioral correlation across sessions and across sites.

We do not assume an adversary with privileged access to the user's own device, nor one capable of compromising the user's local environment. The protections described here concern what is exposed to the web during interaction, not the security of the endpoint from which delegation originates.

The property we seek is unattributability. An external observer should be unable to bind an interaction to the user who delegated it, and should be unable to determine that two distinct interactions were delegated by the same user. This must hold even when the observer can aggregate freely across services and over time.

3 Design Principles

Three principles govern the system.

Delegation over exposure. The user does not act on the web directly. Every interaction with a service is performed by an agent acting on the user's behalf. The user's own identity, environment, and fingerprint never make contact with the service. The interface between the user and the web is the agent, and the agent is disposable.

Disposability over concealment. The system does not attempt to hide a persistent identity more effectively. It declines to maintain a persistent identity at all. Each task is performed under an identity created for that task and destroyed on its completion. There is no long-lived profile to conceal because none is permitted to form.

Multiplicity over singularity. A user is not represented by one well-hidden identity but by many unrelated ones. Work is distributed across a set of agents whose identities are isolated from one another. The coordination that unites them toward the user's intent exists only on the user's side of the system and is never exposed to any observer. To the web, there is no single entity to find. There is a crowd with no discernible center.

4 Architecture

4.1 Overview

Burner consists of three layers. At the base is an isolation layer that provides each agent with a distinct and separable browser environment. Above it is a reasoning layer that translates a user's stated intent into a sequence of actions performed in that environment. Coordinating these is an orchestration layer that decomposes tasks, distributes them across agents, and reconciles their results, while ensuring that the agents remain mutually unlinkable from any external vantage point.

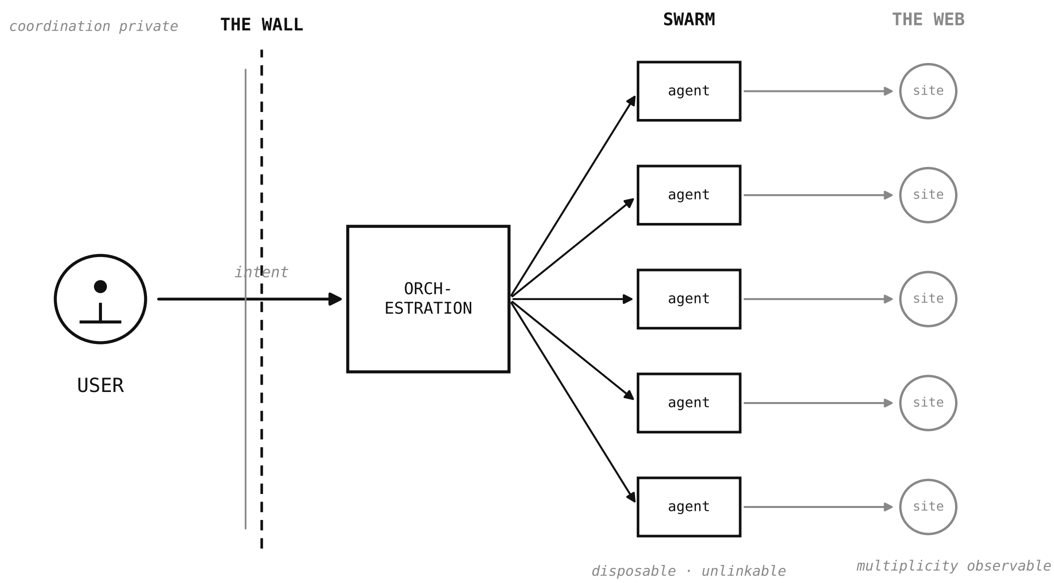

Figure 1: A user delegates intent across the wall to the orchestration layer, which fans the work out to a swarm of disposable, mutually unlinkable agents. Coordination remains private to the user's side; only the multiplicity is observable to the web.

4.2 Isolation Layer

The foundation of the system is a browser environment engineered for autonomous agents rather than adapted from one built for human users. The distinction is material. A browser built for people carries assumptions about persistence, convenience, and continuity that are precisely the properties an attributability adversary exploits. The isolation layer instead treats separation as its primary function.

Each agent operates within its own isolated context, possessing its own browser fingerprint, its own client-side state, and its own apparent device characteristics. No two contexts share a signature. There is no common identifier, persistent store, or behavioral baseline carried between them. Because the contexts share nothing, an observer who characterizes one learns nothing applicable to another. The isolation is not a configuration applied to a shared environment; it is the structure of the environment itself.

4.3 Reasoning Layer

Within an isolated context, an agent must accomplish the task delegated to it. The reasoning layer provides this capability. Given an intent expressed in natural language, the agent forms a plan, then executes it against the live web by perceiving each page as it is encountered and deciding the next action in context.

This live, in-context reasoning distinguishes the agent from conventional automation. A recorded script presumes a fixed sequence of interactions and fails when a page deviates from expectation. An agent that reasons over each page as it appears adapts to variation, recovers from unexpected states, and operates on the dynamic, inconsistent, and frequently hostile pages that comprise the real web. The agent reads, decides, and acts in a continuous loop until the task is complete or it determines that completion is not possible, at which point it reports its terminal state.

4.4 Orchestration Layer

Many tasks exceed the scope of a single interaction. For these, the orchestration layer decomposes the task into parts and assigns each to its own agent. The agents execute concurrently, each within its own isolated context, none aware of the others.

The orchestration that binds these agents to a common purpose exists entirely within the user's trust boundary. It is never represented to any service the agents interact with. From the perspective of an external observer, the agents are unrelated: they present distinct fingerprints, originate from distinct apparent locations, and exhibit no shared identifier or coordinated behavioral signature. The relationship among them is visible only on the user's side of the system, where the results are reconciled into the single outcome the user requested. The coordination is private; only the multiplicity is observable.

This is the structural realization of the multiplicity principle. The system is not a single agent that takes care to remain hidden. It is a swarm with no exposed center, in which the absence of a discoverable coordinating identity is a property of the architecture rather than a matter of operational care.

5 Identity Lifecycle

The unit of identity in Burner is the task, not the user and not the session in any durable sense. An identity is instantiated for a task, exists only while that task is performed, and is destroyed when it concludes.

Instantiation. When a task begins, a fresh identity is created: a new fingerprint, a new context, a new apparent device, sharing nothing with any prior identity. The identity has no history, because it has no past. It is, to any service it encounters, a participant never seen before and never to be seen again.

Operation. During the task, the identity accrues exactly the state required to complete the work and nothing beyond it. No state is retained speculatively against possible future use, because retained state is the substrate from which profiles are built.

Destruction. On completion, the identity is destroyed. The context is torn down, its state is discarded, and nothing is preserved. Completion and erasure are a single event. There is no archive, no fallback copy, and no record kept against the possibility that it might later be wanted. The trail that ordinary interaction would leave is not cleaned up after the fact; it is never permitted to outlive the task that produced it.

The consequence is that the records on which attribution depends do not accumulate. A persistent profile cannot form, because the persistent identity it would attach to is never allowed to exist.

6 Properties

The system provides several properties against the threat model of Section 2.

Non-attribution to the user. Because the user never interacts with a service directly, and because every interaction is conducted by a disposable identity sharing nothing with the user's own environment, no interaction can be bound to the user by the means available to the adversary.

Non-linkability across interactions. Because each task is performed under an identity that shares no fingerprint, state, or behavioral baseline with any other, an observer cannot determine that two interactions originate from the same source. This holds across services and over time, defeating correlation by aggregation.

Absence of a persistent profile. Because identities are destroyed on task completion and no state is retained, there is no durable record against which a profile can be assembled. The adversary's primary instrument, the accumulation of behavior over a stable identity, is denied its precondition.

No exposed center. Because the coordination among agents exists only within the user's trust boundary, an observer of the swarm sees unrelated participants and no coordinating entity. There is no single identity whose discovery would compromise the others.

7 Limitations

We state the boundaries of the approach plainly, as any honest account must.

The protections concern what is exposed to the web during interaction. They do not extend to the security of the user's own endpoint. A compromised device from which delegation originates is outside the threat model and outside what the system can defend.

Unattributability is a property relative to the adversary defined in Section 2. An adversary with capabilities beyond that model, including privileged access to infrastructure or to the user's trust boundary, is not addressed here. No claim is made of protection against such an adversary.

The system reduces the correlation available to an observer; it does not render correlation categorically impossible under all conceivable analyses. Behavioral signals not anticipated by the isolation layer may in principle offer some residual correlative power. The design seeks to minimize this surface, and the absence of shared identifiers across contexts removes the most powerful correlative instruments, but no system that interacts with the web can claim a complete absence of observable signal.

Finally, the agent operates on the web as it is. Services evolve, and an interaction model is never static. The system's effectiveness is a function of ongoing engineering against a changing environment, not a fixed guarantee established once.

8 Conclusion

The web identifies by default, and the prevailing privacy tools accept that default while attempting to mitigate its consequences. Burner declines the default itself. By delegating every interaction to a disposable agent, destroying each identity on completion, and distributing work across a swarm whose coordination is never exposed, the system represents a user online not as a single observable entity but as a continuous supply of unrelated, short-lived strangers.

The objective is not concealment of a persistent self. It is the refusal to present a persistent self to be observed. The agents act. The user remains behind the interface. What the web is given to record is, in the end, nothing that leads back.

---

complies with the terms of the services they interact with and with applicable law in their jurisdiction.